# Time-Dependent ROC

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## Introduction

Time-dependent ROC extends classical ROC analysis to right-censored survival data. At each chosen evaluation time $t$, the analysis treats subjects observed to fail before $t$ as cases and subjects known to survive past $t$ as controls; subjects censored before $t$ are reweighted via inverse-probability-of-censoring weights. The resulting time-specific AUC summarises how well a continuous risk score (such as a Cox linear predictor) discriminates failures from survivors at $t$. It is the natural extension of the C-index to "what is the AUC at exactly one year" reporting.

## Prerequisites

A working understanding of ROC curves, the C-index, Cox regression linear predictors, and inverse-probability-of-censoring weighting.

## Theory

Two definitions of cases and controls compete:

- **Incident / dynamic** ($I/D$): cases are subjects who fail at time $t$ exactly; controls are those still at risk past $t$. Useful for hazard-style discrimination.
- **Cumulative / dynamic** ($C/D$): cases are subjects who fail at any time $\leq t$; controls are those still at risk past $t$. The more common choice for clinical risk-score reporting.

Inverse-probability-of-censoring weights compensate for subjects censored before the comparison can be made; the censoring distribution is typically estimated by Kaplan-Meier on the censoring times. The result is a time-dependent AUC at each requested $t$ with bootstrap or analytical confidence intervals.

## Assumptions

Non-informative censoring (or correctly conditional on covariates if `weighting = "cox"` or `"aalen"` is used) and a continuous risk marker that is meaningful as a ranking score.

## R Implementation

```r
library(timeROC); library(survival)

data(lung)
fit <- coxph(Surv(time, status) ~ age + sex + ph.ecog, data = lung)
lp <- predict(fit, type = "lp")

tROC <- timeROC(T = lung$time, delta = lung$status, marker = lp,
                times = c(180, 365, 730), cause = 1,
                weighting = "marginal")
tROC
plot(tROC, time = 365, col = "steelblue")
```

## Output & Results

`timeROC` returns time-specific AUCs with their standard errors at each requested horizon, along with empirical sensitivity-specificity pairs that can be plotted as a ROC curve. The plot shows the ROC curve at one selected time; running over multiple times reveals how discrimination evolves through follow-up.

## Interpretation

A reporting sentence: "The Cox linear predictor's time-dependent AUC was 0.72 (95 % CI 0.65 to 0.79) at one year and 0.69 (95 % CI 0.62 to 0.76) at two years; discrimination remained stable across the two-year window relevant to clinical decision-making." Always report AUCs at clinically meaningful horizons, not just the global C-index, because discrimination can change substantially across follow-up.

## Practical Tips

- Report AUCs at the horizons that drive clinical decisions; a five-year AUC is irrelevant for a six-month treatment decision and vice versa.
- The cumulative/dynamic definition is more common in clinical reporting; incident/dynamic is more useful when modelling hazards directly.
- `timeROC()` supports competing risks via the `cause` argument; the same workflow applies with the failure of interest specified.
- AUC alone does not capture calibration; pair time-dependent AUCs with calibration plots and Brier scores at the same times.
- For external validation, apply the development-cohort risk score to the validation cohort and compute time-dependent AUCs there; in-sample AUCs are optimistic.
- Use `weighting = "cox"` or `weighting = "aalen"` when censoring depends on covariates; the marginal weighting assumes censoring is uninformative given baseline covariates.

## R Packages Used

`timeROC` for `timeROC()` with marginal, Cox, and Aalen-based weighting; `risksetROC` for an alternative kernel-smoothed implementation; `pec` for compatible prediction-error curves; `survAUC` for additional AUC variants including Uno's C-statistic.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation